

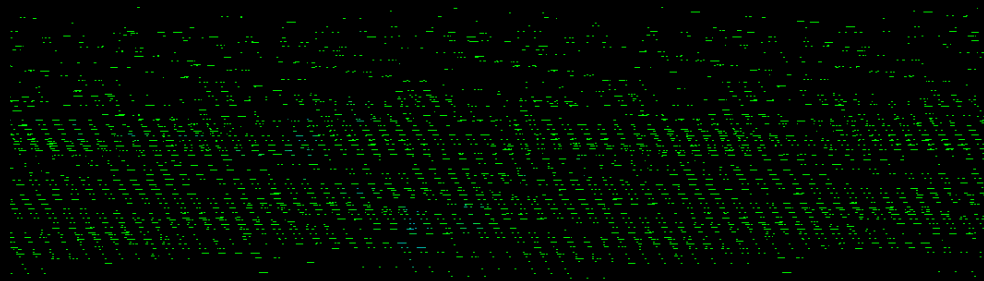
vaadin}>

Your Backend Deserves a Face

A Hands-on Workshop for
Spring Boot & Vaadin



Sebastian Kuehnau
Vaadin Enthusiast



Disclaimer

Who is talking?

Sebastian Kuehnau

+20y in web- & java development

+10y at Vaadin

several technical roles

now developer advocate

Off-work mostly outdoors;



From zero to full-stack Java web app

What we'll build



Browse talks in a grid



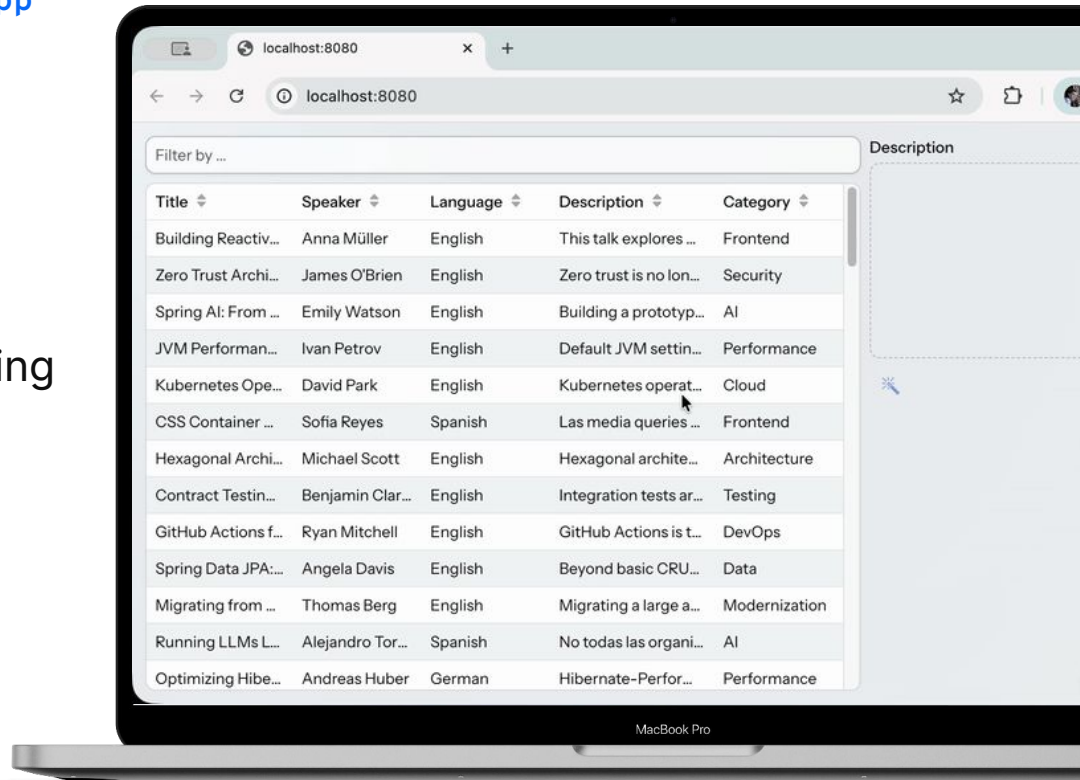
Lazy loading, sorting & filtering



AI-powered summaries



Browserless UI testing





*This slide exists because
the setup takes a moment.*

GOTO: 📖

[tinyurl.com/
springio26-vaadin](https://tinyurl.com/springio26-vaadin)

01

Vaadin Birds-Eye View

Who's using it?

How is it different?

How does it work?

02

Live Coding

The full-stack developer's dilemma

Why WebDev is hard with Java?



Outdated tech or switching stacks



Modern web frameworks require a different language and toolchain



Teams work in silos – backend vs. frontend, Java vs. JS

Fast and pragmatic



Building Web Apps with Java only

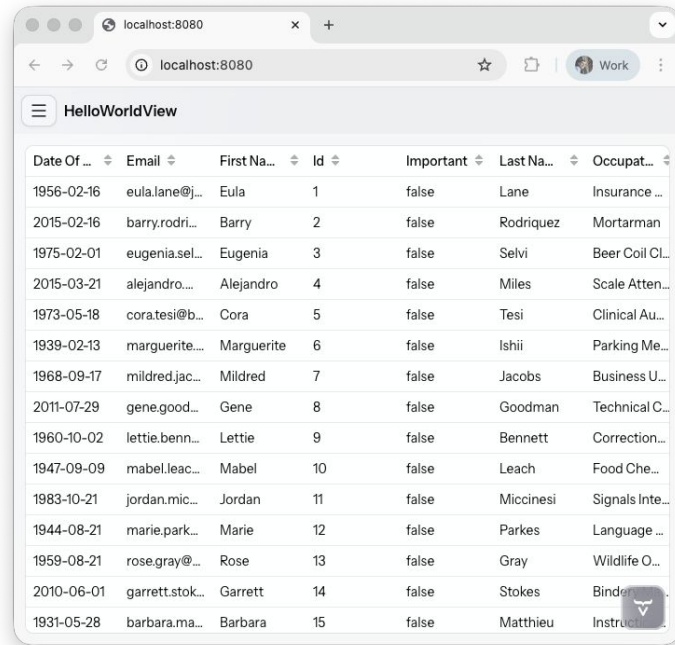
Vaadin

```
@Route(value = "")
public class HelloWorldView extends HorizontalLayout {

    public HelloWorldView(PersonRepository repository) {

        var grid = new Grid<>(Person.class);
        grid.setItems(repository.findAll());

        add(grid);
    }
}
```



Date Of ...	Email	First Na...	Id	Important	Last Na...	Occupat...
1956-02-16	eula.lane@j...	Eula	1	false	Lane	Insurance ...
2015-02-16	barry.rodri...	Barry	2	false	Rodriquez	Mortarman
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1983-10-21	jordan.mic...	Jordan	11	false	Miccinesi	Signals Inte...
1944-08-21	marie.park...	Marie	12	false	Parkes	Language ...
1959-08-21	rose.gray@...	Rose	13	false	Gray	Wildlife O...
2010-06-01	garrett.stok...	Garrett	14	false	Stokes	Binder
1931-05-28	barbara.ma...	Barbara	15	false	Matthieu	Instructio...

..., but how does
my **Java** code get
to the **Browser**?



Architecture

```
▼<vaadin-horizontal-layout theme="spacing" has-st
  ▶ #shadow-root (open)
  ▶ <vaadin-text-field placeholder="Your name">⋮
  ▶ <vaadin-button tabindex="0" role="button" last
</vaadin-horizontal-layout>
```

```
var name = new TextField("Your name");
var button = new Button("Say hello", event ->
    Notification.show("Hello " + name.getValue()));
add(new HorizontalLayout(name, button));
```

Architecture

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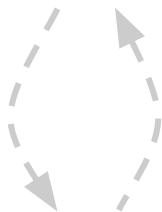
*Vaadin UI Components
Web Components*

UI-Code (generated)
HTML + JS

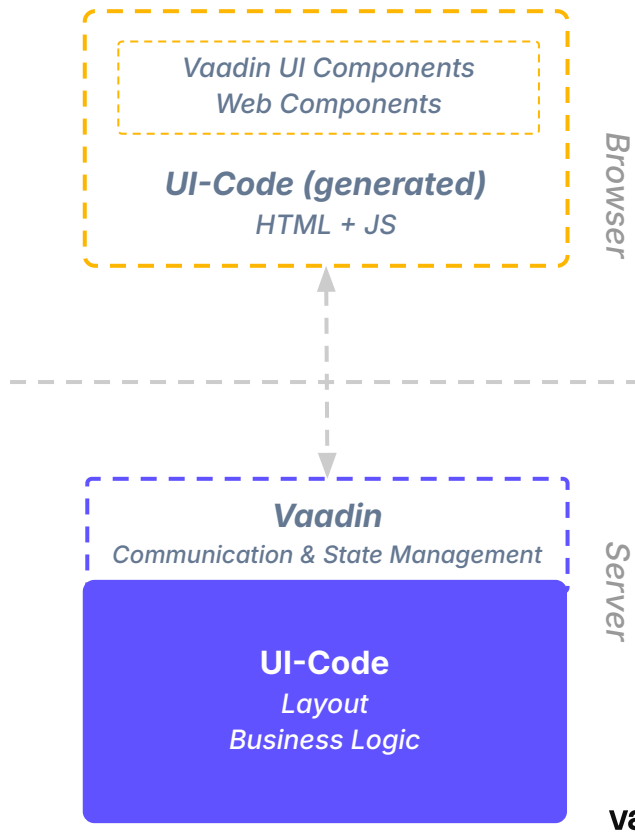
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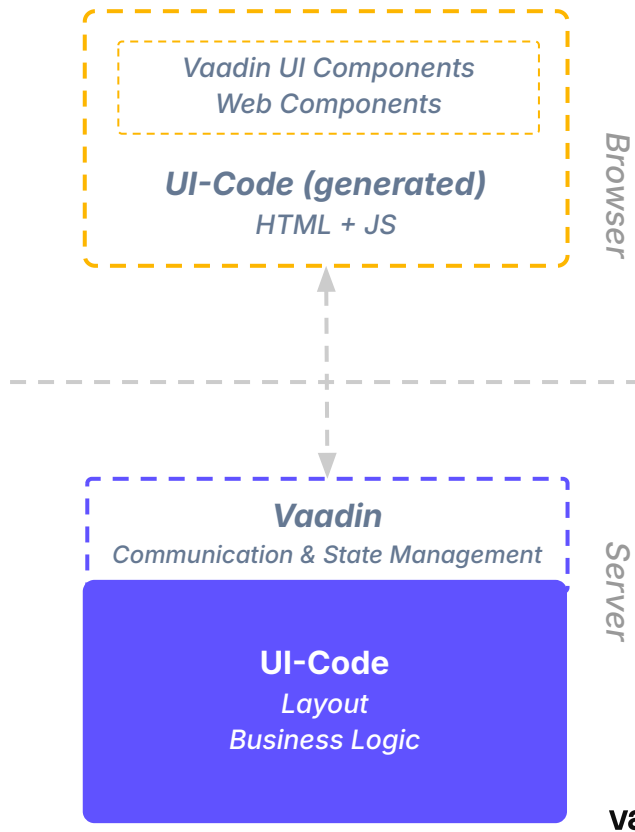


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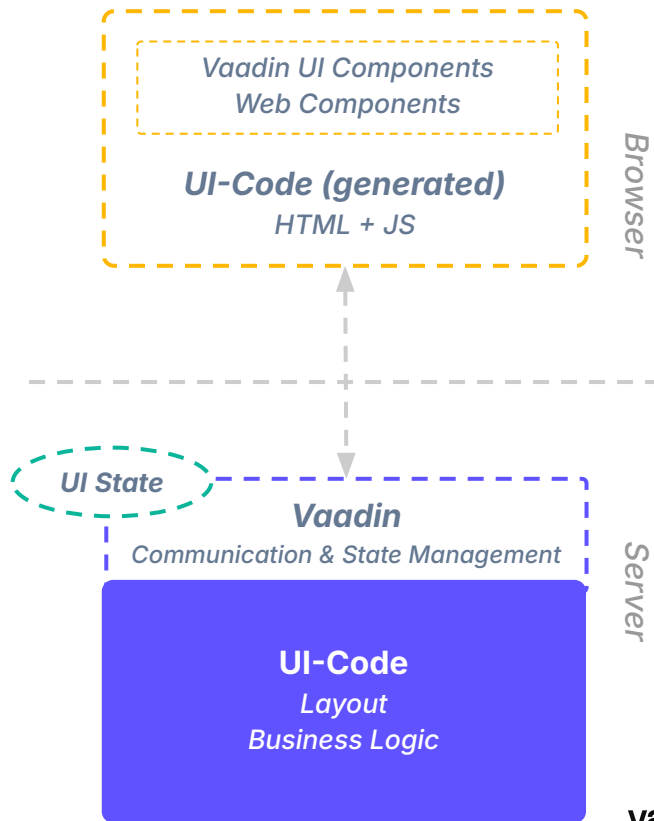


Architecture

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Architecture

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  </vaadin-horizontal-layout>
```

HTTP[S] Post

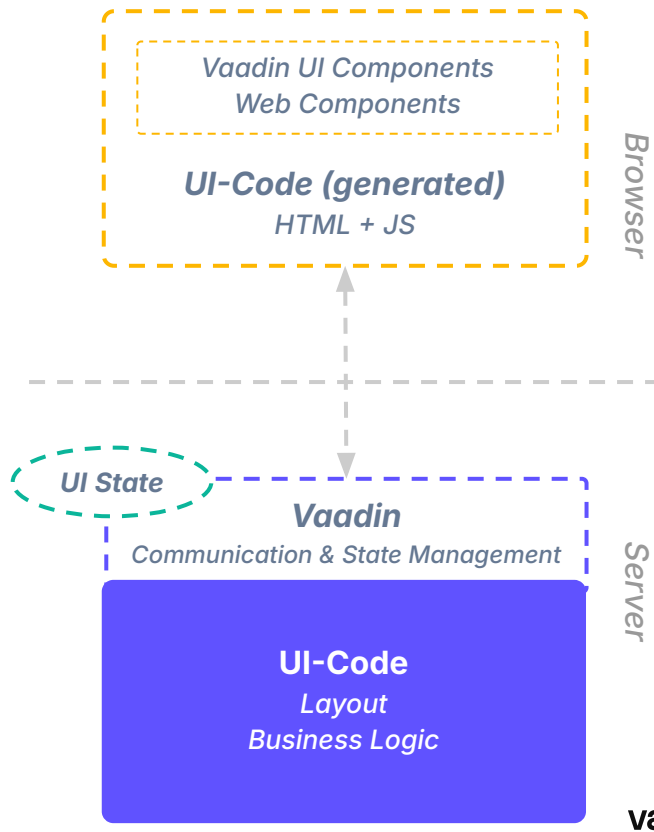
with clientId &
csrfToken



JSON Response

with clientId +1,
syncId + 1 & csrfToken

```
var name = new TextField("Your name");
var button = new Button("Say hello", event ->
    Notification.show("Hello " + name.getValue()));
add(new HorizontalLayout(name, button));
```



The Vaadin way }>

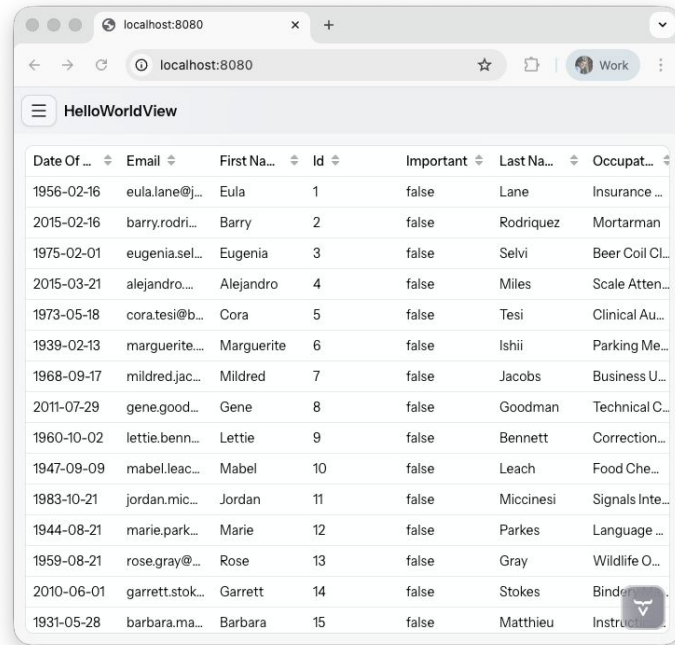
Just Java

```
@Route(value = "")
public class HelloWorldView extends HorizontalLayout {

    public HelloWorldView(PersonRepository repository) {

        var grid = new Grid<>(Person.class);
        grid.setItems(repository.findAll());

        add(grid);
    }
}
```

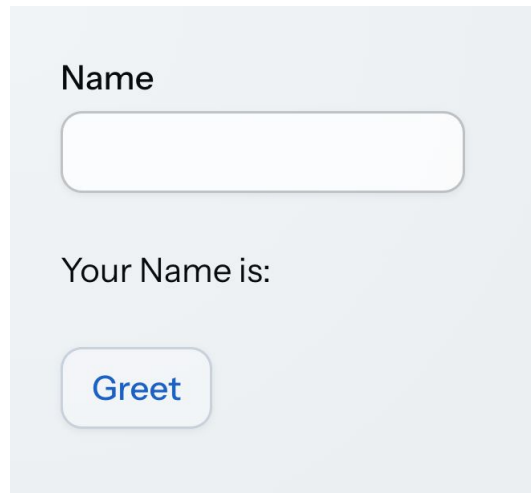


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2010-06-01	garrett.stok...	Garrett	14	false	Stokes	Binder
1931-05-28	barbara.ma...	Barbara	15	false	Matthieu	Instructio...

Compose your UI like you compose your objects

Components

```
var name = new TextField("Name");  
var greeting = new Paragraph("Your Name is:");  
var sayHello = new Button("Greet");  
  
var rootLayout = new VerticalLayout(  
    name,  
    greeting,  
    sayHello);
```



The screenshot shows a light gray rectangular area representing a web page. At the top, the word "Name" is displayed in a dark gray font. Below it is a white text input field with rounded corners and a thin gray border. Further down, the text "Your Name is:" is shown in a dark gray font. At the bottom of the form is a button with rounded corners, a thin gray border, and the word "Greet" in a blue font.

Nest them like your any other objects

Components

```
var name = new TextField("Name");  
var greeting = new Paragraph("Your Name is:");  
var sayHello = new Button("Greet");  
  
var rootLayout = new VerticalLayout(  
    new HorizontalLayout(name, greeting),  
    sayHello);
```

Name

Your Name is:

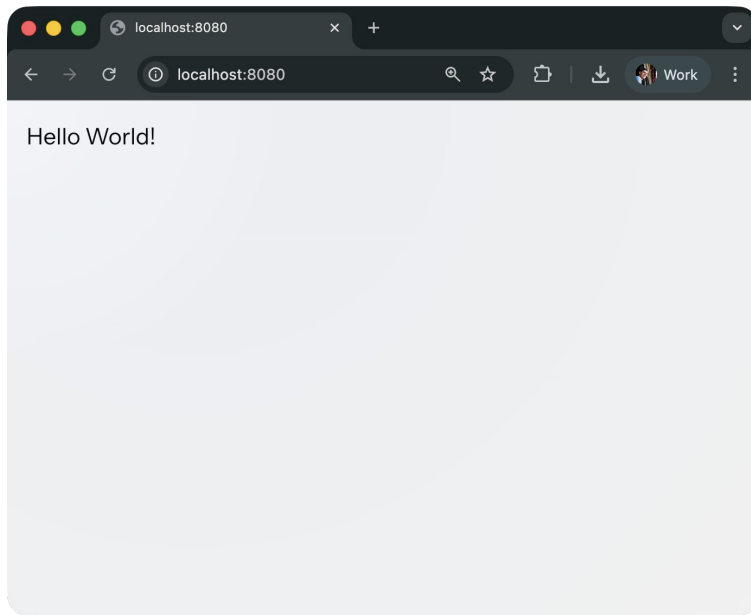
Greet

Your first route

Navigation

```
@Route("")
public class HelloWorldView extends
    VerticalLayout {

    public HelloWorldView() {
        add("Hello World!");
    }
}
```

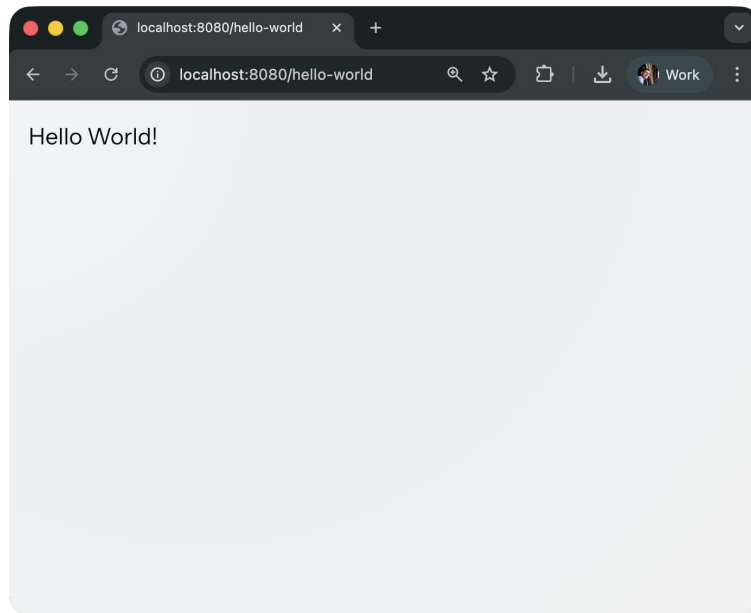


Named routes

Navigation

```
@Route("hello-world")
public class HelloWorldView extends
    VerticalLayout {

    public HelloWorldView() {
        add("Hello World!");
    }
}
```



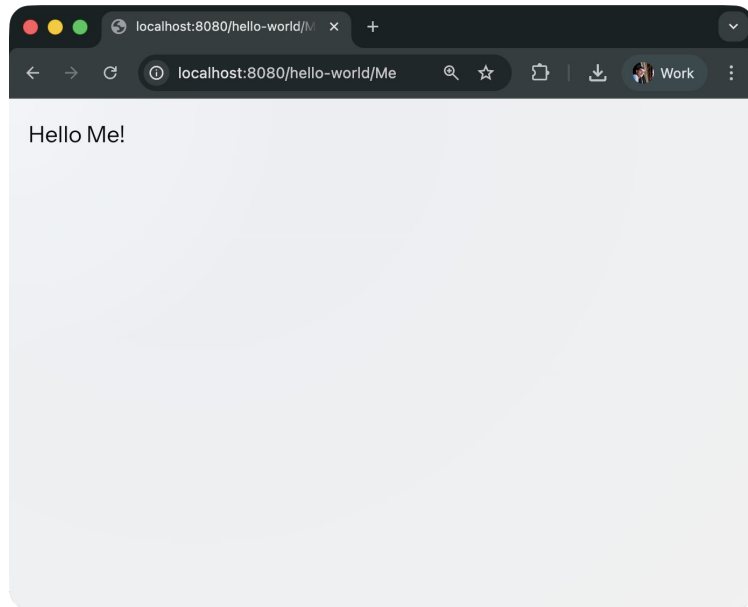
Routes with parameters

Navigation

```
@Route("hello-world")
public class HelloWorldView ...
    implements HasUrlParameter<String> {

    public HelloWorldView() { }

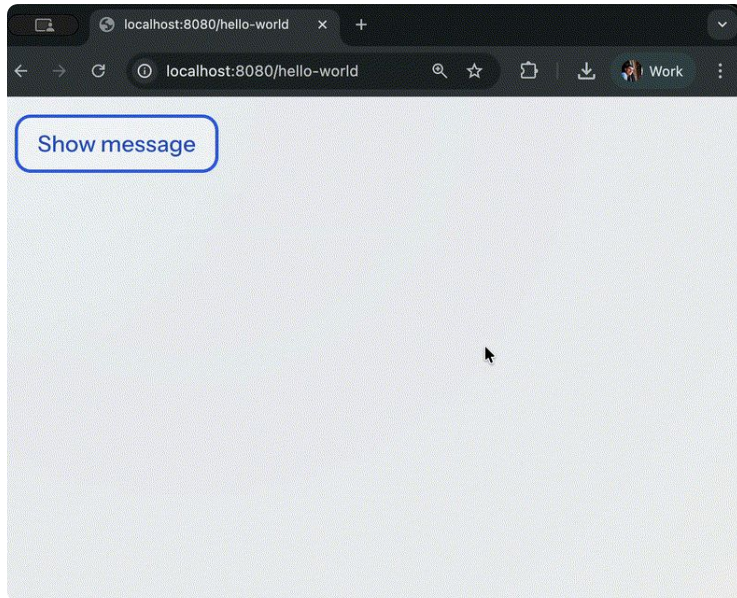
    @Override
    public void setParameter(BeforeEvent event,
        String parameter) {
        add("Hello " + parameter + "!");
    }
}
```



Browser clicks – Java reacts

Event Handling

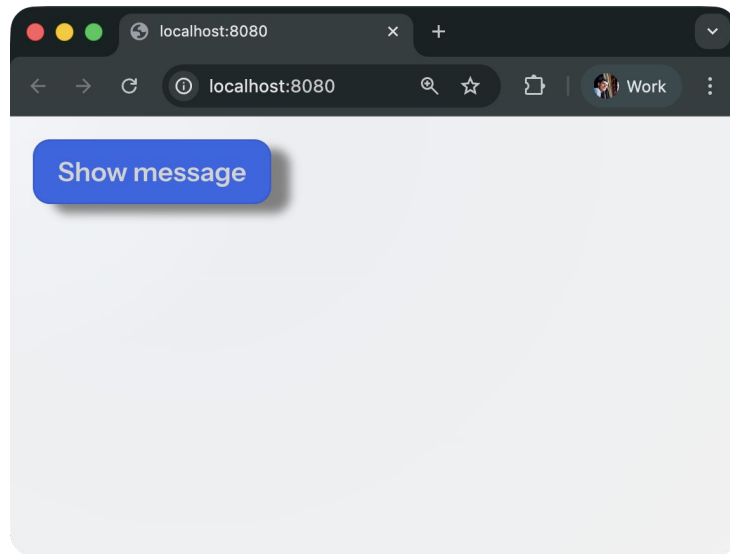
```
var button = new Button("Show message");  
button.addClickListener(event →  
    Notification.show("Hello World!"));  
add(button);
```



Style via Java API or CSS

Theming & Styling

```
var button = new Button("Show message");
button.addThemeVariants(ButtonVariant.PRIMARY);
button.getStyle().set("box-shadow",
    "10px 5px 5px gray");
button.addClassName("dark-font");
//.dark-font { color:gray }
add(button);
```



Apache 2.0

Same as Spring + No license fees +
commercial friendly + allow code changes

github.com/vaadin

(Sources, Issues, Feedback)

vaadin.com

(Docs, Forum, Webinars, Trainings, Addons)

01

Vaadin Birds-Eye View

02

Live Demo

Setup your Project

Views and Components

Data Grids with Lazy Loading

Benefit from AI in your UI

Where to go from here?

Get started with Vaadin



Slides
Demo Project

seb@vaadin.com

[github.com/
SebastianKuehnau/
talk-tracker/tree/
events/springio-26/
solution](https://github.com/SebastianKuehnau/talk-tracker/tree/events/springio-26/solution)

Meet us at
Vaadin Create 2026
in Barcelona
Oct 27-28, 2026

Discount Code:
CREATESPRING30

[vaadin.com/
vaadin-create-2026](https://vaadin.com/vaadin-create-2026)

vaadin}>

Kiitos paljon!

Feel free to reach out

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linkedin.com/in/sebastian-kuehnau/

github.com/sebastianKuehnau

vaadin.com/blog